

كلية الطب – جامعة العميد

الحملة التوعوية حول الحصول على

الاعتمادية

الرؤية

التميز الفعال من اجل رفد المجتمع بأطباء اكفاء في الخدمة الطبية والبحث العلمي

الرسالة

توفير بيئة جامعية تعمل بمنهج تعليمي مواكب للتطور
اطباء ذوي كفاءة ومستوى علمي قادرين على تقديم الخدمة المتميزة



الاهداف

- اعداد اطباء بكفاءة عالية
- التأكيد على مبادئ الاخلاق الطبية
- نشر مفاهيم العمل كفريق واحد
- العمل على حل المشكلات الصحية
- تطبيق مخرجات البحوث المتقدمة
- تعزيز عملية البحث العلمي
- تبني مفاهيم التعلم المهني المستمر



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Introduction- Pharmacodynamics

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Pharmacodynamics

- + Pharmacodynamics describes the actions of a drug on the body.
- + Drugs act as signals, and their target (like receptors) act as signal detectors

Drug + Receptor $\leftrightarrow$ Drug-receptor complex $\rightarrow$
Biological effect.

Target for drug action

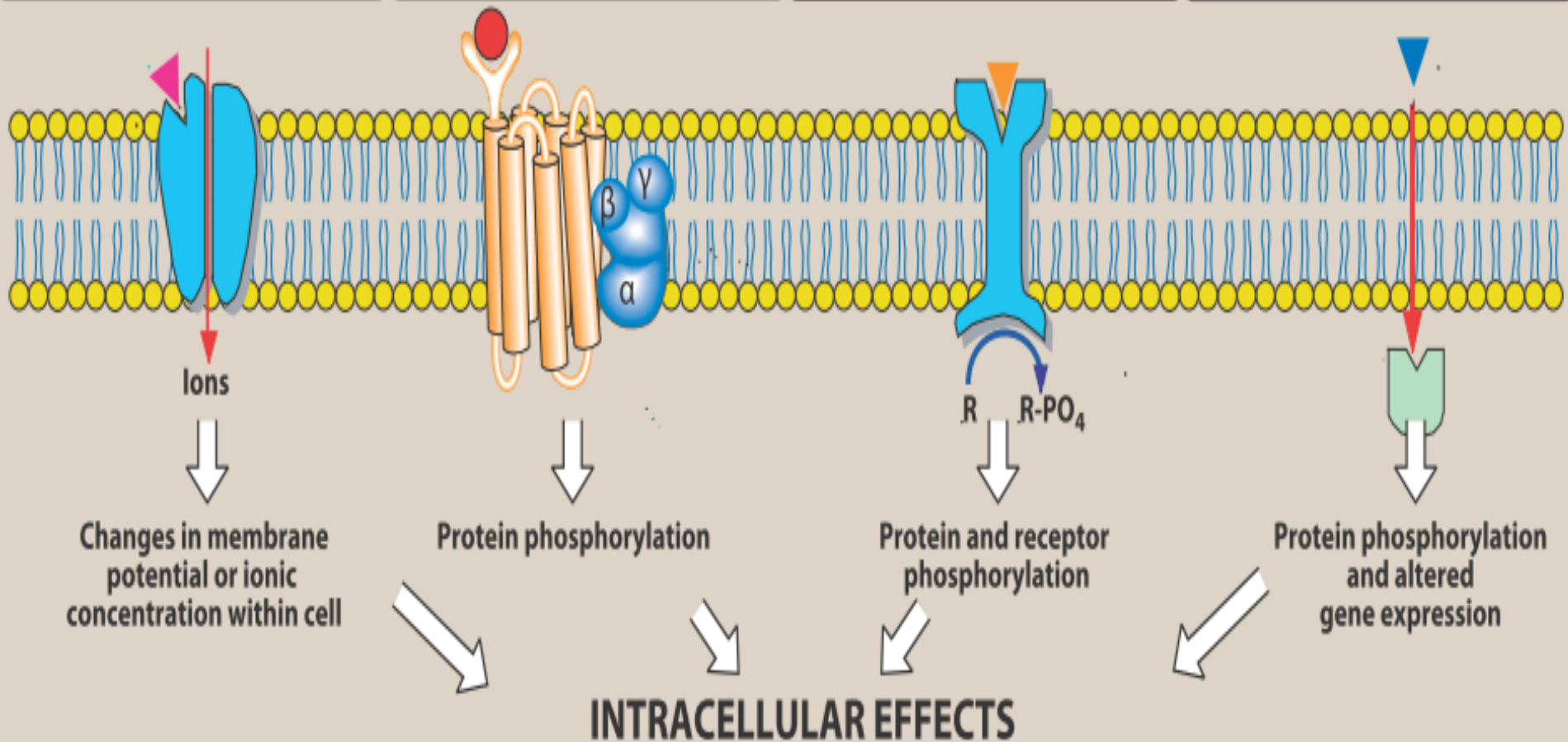
1. **Receptors** (most common and important)
2. **Enzymes:** like cyclo- oxygenase inhibitors like aspirin.
3. **Transport (carrier) protein system:** like hemicholinium blocks choline re-uptake.
4. **Ion channels:** like blocking of Na⁺ channels by lidocaine

Receptor

- It is (usually protein) any cellular macromolecule to which a drug binds to initiate its effects.

Types of receptors

A Ligand-gated ion channels <u>Example:</u> Cholinergic nicotinic receptors	B G protein-coupled receptors <u>Example:</u> α and β adrenoceptors	C Enzyme-linked receptors <u>Example:</u> Insulin receptors	D Intracellular receptors <u>Example:</u> Steroid receptors
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1- Transmembrane ligand-gated ion channels

They are responsible for regulation of the flow of ions across cell membranes.

- Response to these receptors is very rapid, enduring for only a few milliseconds.

E.g.

- Nicotinic receptor for acetyl choline
- γ -aminobutyric acid (GABA) receptor by GABA

2- Transmembrane G protein-coupled receptors(GPCR)

- They comprise a single α helical peptide that has seven membrane-spanning regions.
- The extracellular domain contains the ligand-binding area.
- Intra-cellularly, these receptors are linked to a cytosolic G protein effector (e.g. second messenger)
- Stimulation of these receptors results in responses that last several seconds to minutes.
- They are the most abundant type of receptors.

Second messengers

These are essential in conducting and amplifying signals coming from G protein–coupled receptors.

- ➡ Gs → activation of adenylyl cyclase → production of cyclic adenosine monophosphate (cAMP)—a second messenger (regulates protein phosphorylation).
- ➡ Gi → inhibition of adenylyl cyclase → decrease cAMP
- ➡ Gq → activation of Phospholipase C → inositol-1,4,5-trisphosphate (IP3) and diacyl-glycerol (DAG). IP3 is responsible for the regulation of intracellular free calcium concentrations. DAG activates several enzymes such as protein kinase C.

3- Enzyme-linked receptors

- They consist of a protein that spans the membrane once and may form dimers or multi subunit complexes.
- These receptors also have cytosolic enzyme (e.g. tyrosine kinase) activity as an integral component of their structure and function.
- Duration of responses to stimulation of these receptors is minutes to hours.
- E.g.: (insulin, epidermal growth factor, platelet-derived growth factor, atrial natriuretic peptide , and others).

4- Intracellular receptors (nuclear receptors)

- The receptor is entirely intracellular, and, therefore, the ligand must be highly lipid soluble to diffuse into the cell to interact with the receptor.
- The primary targets of these ligand-receptor complexes are transcription factors for DNA and RNA.
- Cellular responses are not observed until considerable time has elapsed (30 minutes or more), and the duration of the response (hours to days)
- e.g. steroid hormones (cellular) , thyroid hormone (nuclear), vitamin D and A.

Some characteristics of signal transduction

- 1)** *The ability to amplify small signals duration and intensity:* in which the only a fraction of the total receptors for a specific ligand may need to elicit a maximal response from a cell.
- Systems that exhibit this behavior are said to have *spare receptors.* About 99 %of Insulin receptors are “spare.”

Some characteristics of signal transduction

2) *protection the cell from excessive stimulation (Desensitization and down-regulation of receptors):* Several mechanisms have evolved to protect a cell from excessive stimulation.

When repeated administration of a drug results in a diminished effect, the phenomenon is called *tachyphylaxis*.

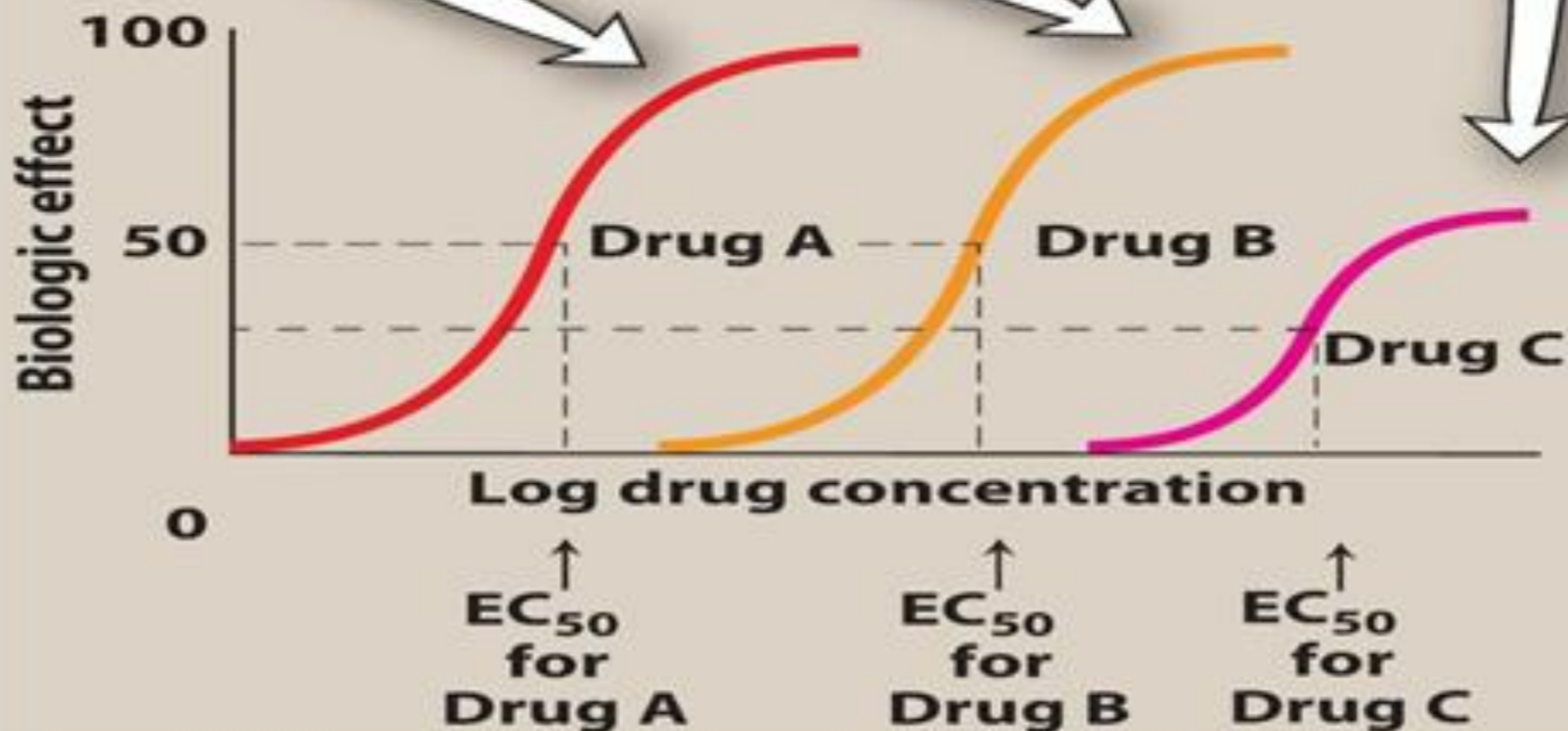
The receptor becomes desensitized to the action of the drug and require a time (rest period) following stimulation before they can be activated again. During this recovery phase they are said to be *“refractory” or “unresponsive.”*

Important terms

- Potency: the amount of drug necessary to produce an effect of a given strength. *((The concentration of drug producing an effect that is 50 percent of the maximum is used to determine potency and is commonly designated as the EC_{50})).*
- Efficacy: This is the ability of a drug to give a response when it interacts with a receptor.
- The maximal response (E max), or efficacy, is more important than drug potency.

Drug A is more potent than Drug B, but both show the same efficacy.

Drug C shows lower potency and lower efficacy than Drugs A and B.



Drug-receptor relationship

- *AGONISTS*

An agonist binds to a receptor and produces a biologic response.

Types of agonists

1- Full agonists:

- The drug binds to a receptor and produces a maximal biologic response that mimics the response to the endogenous ligand like phenylephrine is an agonist at α 1-adrenoceptors.
- In general, a full agonist has a strong *affinity* for its receptor and maximum *efficacy*.

Types of agonists-continue

2- Partial agonists:

- Have efficacies (intrinsic activities) greater than zero but less than that of a full agonist.
- Partial agonist may have an affinity that is greater than, less than, or equivalent to that of a full agonist.
- *A unique feature* of these drugs is that, under appropriate conditions, a partial agonist may act as an antagonist of a full agonist e.g. aripiprazole, an atypical neuroleptic agent

Types of agonists-continue

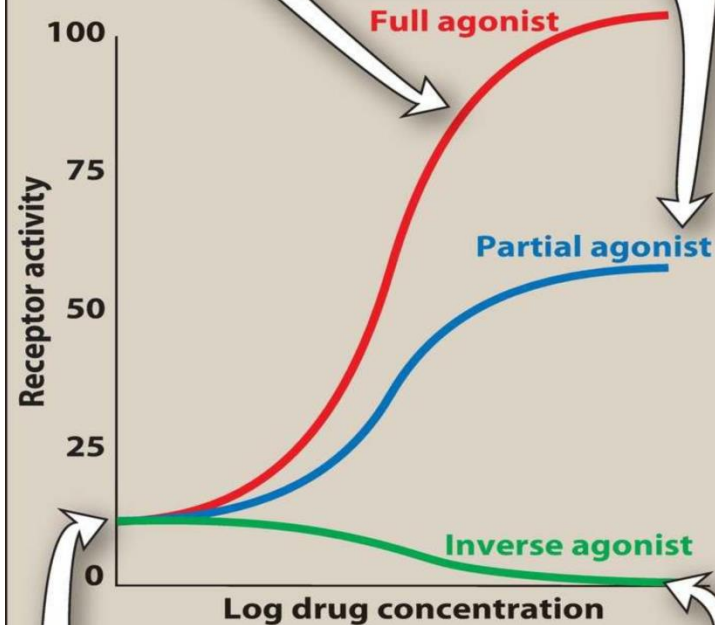
3- Inverse agonists:

- It is an agent that binds to the same receptor as an agonist but induces a pharmacological response opposite to that agonist.
- An inverse agonist has $< 0\%$ (i.e., negative) efficacy.

e.g. Betacarbolines on BDZ receptors.

A full agonist produces complete activation of a receptor at high drug concentrations.

Partial agonist binding results in less than 100% activation, even at very high concentrations.



Inverse agonists produce a response below the base-line response measured in the absence of drug.

In this example, a portion of the receptors show constitutive activity (without stimulation), such that 12% of maximal response is seen.

ANTAGONISTS

- Antagonists are drugs that decrease or oppose the actions of another drug or endogenous ligand i.e. an antagonist has no effect if an agonist is not present.

Types of antagonism

1-Pharmacological antagonism

A. Competitive antagonists: If both the antagonist and the agonist bind to the same site on the receptor equally, they are said to be “competitive.” Like histamine and loratidine.

B- Irreversible antagonists:

- The effects of irreversible antagonists, by contrast to competitive one, cannot be overcome by adding more agonist.
- This occurs due to either by bind covalently or by binding to ("allosteric site").

2-Functional and chemical antagonism

- Functional antagonist "physiologic antagonism"

acts at a completely separate receptor, initiating effects that are functionally opposite those of the agonist.

e.g. H₁ histamine receptors activation causes contraction and narrowing of the bronchial tree. Epinephrine is an agonist at β_2 -adrenoceptors on bronchial smooth muscle, which causes the muscles to actively relax.

- Chemical antagonist

prevents the actions of an agonist by modifying or sequestering the agonist so that it is incapable of binding to and activating its receptor. For example, protamine sulfate is a chemical antagonist for heparin.

Therapeutic index

$$\text{Therapeutic index} = \text{TD}_{50} / \text{ED}_{50}$$

- ✓ The therapeutic index is a measure of a drug's safety, because a larger value indicates a wide margin between doses that are effective and doses that are toxic.
- Warfarin (example of a drug with a small therapeutic index)
- Penicillin (example of a drug with a large therapeutic index)